PETER O'BRIEN

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Education

Ph.D. in Agricultural and Resource Economics, University of California, Davis — Expected 2027
B.S. in Economics, Minor in Mathematics, *summa cum laude*, University of Maine — 2021

Fields of Interest

Environmental Economics, International Trade, and Agricultural Economics

References

Katrina Jessoe **(Co-chair)**
Associate Professor
Agricultural and Resource Economics
UC Davis
kkjessoe@ucdavis.edu

James Sayre **(Co-chair)**
Assistant Professor
Agricultural and Resource Economics
UC Davis
jsayre@ucdavis.edu

Ishan Nath
Assistant Professor
Harvard Kennedy School
Harvard University
ishannath@hks.harvard.edu

Research in Progress

Feeding the Export Boom: Technological Change and Land Use in Brazil's Beef Industry (Job Market Paper)

Abstract: This paper studies whether increased export market exposure induces agricultural technological change and how that response affects land use. I examine Brazil's beef export boom to China, exploiting staggered plant authorizations combined with a novel panel of feedlot development that I construct using satellite imagery and deep learning. I first document that beef-processing plants increasingly source younger cattle and draw more heavily from regions with feedlots after gaining access to the Chinese export market. Exploiting variation in the timing of plant authorizations and municipalities' exposure to newly authorized plants, I find that access to the Chinese market substantially increased feedlot entry and confined-finishing capacity. My back-of-the-envelope estimates suggest that replacing this China-induced feedlot finishing with pasture finishing would require approximately 1.29 million additional hectares of pasture, which exceeds the observed total expansion of Brazil's pasture area between 2004 and 2024, and would imply 169 $MtCO_2$ of potential carbon stock losses. The results show how trade can affect environmental resource use by inducing changes in production technology.

Does Eating Local Reduce Emissions?
with Levi Crews, Ishan Nath, Milan Quentel, and James Sayre

Publications

Do California's Cattle Emit Less Carbon Than We Think? Published in *ARE Update* in 2025 with James Sayre.

Research Experience

Graduate Student Researcher — 2024 – 2025
with James Sayre

Graduate Student Researcher — 2023 – 2024
with Michael Carter

Teaching Experience

Graduate Teaching Assistant, UC Davis

Econometrics (M.S. level) with Timothy Beatty	Fall 2026
Operations Science with Bulat Gafarov	Spring 2026
Agricultural Markets with Timothy Beatty	Winter 2025
Economics of Business Regulation with Katrina Jessoe	Fall 2025
Intermediate Microeconomics with Kevin Novan	Fall 2022

Scholarships, Honors, and Awards

2026	Agricultural and Resource Economics Summer Research Fellowship, UC Davis
2024–2026	Jastro-Shields Graduate Research Award, UC Davis
2024	Giannini Foundation Mini-Grant (cooperating personnel, with PI James Sayre)
2024	Agricultural and Resource Economics Summer Research Fellowship, UC Davis
2023	Non-Resident Tuition Fellowship, UC Davis
2019	Lavery Scholarship for Excellence in Economics, University of Maine

Professional Activities

Conference and Seminar Presentations

2021 Northeast Agricultural and Resource Economics Association Annual Meeting

Service

2024–2025 UC Davis Agricultural Economics Workshop, Student Coordinator

Media and Outreach

Extreme Reactions to Extreme Events: The Case of Rice Export Restrictions. Ag Data News, with Greg Boudreaux (2024).

Why do we toss 33% of food we produce when people are hungry? The Maine Question Podcast (2020).

Technical Reports

Bid Inflation for Highway and Bridge Projects: Challenges & Solutions. Prepared for the Maine Department of Transportation (2022).

Other Experience

Research Analyst, Margaret Chase Smith Policy Center 2022

(Updated October 13, 2026)